

Cellular Biophysics (BIO516) – Quiz 1

Marks: 20, Time: 1 hr

Date: 29/01/2026

Instructions to candidates

1. This paper consists of three sections: Section A (Multiple Choice: Q1-Q10), Section B (Fill in the Blanks: Q11-Q16), and Section C (Numerical Problems: Q17 & Q18).

2. **Answer all the questions**

3. Use of calculator is permitted.

4. Standard Constants:

- Gas Constant (R) = $8.315 \text{ J}/(\text{mol} \cdot \text{K})$
- Faraday's Constant (F) = $96,485 \text{ C/mol}$
- Temperature (T) = 298 K (unless specified)

Section A

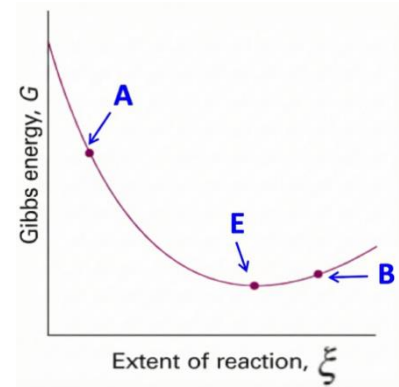
[Marks: 10 X 1]

Q1 What do the points A, B, and E in the figure at the right represent

- 1) A : $\Delta G < 0$, E : $\Delta G = 0$
- 2) B : $\Delta G < 0$, E : $\Delta G = 0$
- 3) A : $\Delta G > 0$, E : $\Delta G = 0$
- 4) B : $\Delta G > 0$, E : $\Delta G = 0$

Which of the combinations is correct?

- A) 1 & 2
- B) 2 & 3
- C) 3 & 4
- D) 1 & 4**



Q2. For a reaction to be spontaneous (exergonic), the change in Gibbs free energy (ΔG) for the system must be:

- A) Positive ($\Delta G > 0$)
- B) Negative ($\Delta G < 0$)**
- C) Zero ($\Delta G = 0$)
- D) Equal to the change in Enthalpy (ΔH)

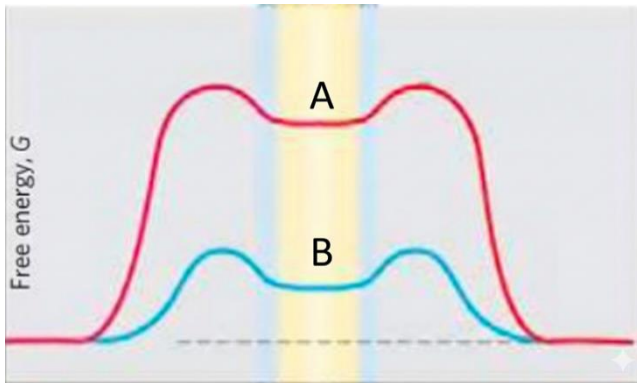
Q3. Choose correct order of hydrophilicity of bonds (ΔEN in parentheses)

- A) N-H (0.84) < S=O (0.86) < C=O (0.89) < O-H (1.24)
- B) C=O (0.89) < N-H (0.84) < S=O (0.86) < O-H (1.24)
- C) C=O (0.89) < N-H (0.84) < O-H (1.24) < S=O (0.86)**
- D) N-H (0.84) < C=O (0.89) < O-H (1.24) < S=O (0.86)

Q4. Which component of the cell membrane is primarily responsible for preventing the clumping of red blood cells by conferring a negative charge?

- A) Phosphatidylcholine
- B) Cholesterol
- C) Glycophorin A**
- D) Spectrin

Q5 The following free energy graphs A & B for the transport of charged/polar solute across the plasma membrane represent;



- 1) A represents Simple diffusion with transporter
- 2) B represents Diffusion with transporter
- 3) A represents Simple diffusion without transporter
- 4) B represents Diffusion without transporter

Choose the following combination of statements

- A) 1 & 2
- B) 1 & 4
- C) 2 & 3**
- D) 3 & 4

Q6. According to the fluid mosaic model, which motion is thermodynamically unfavorable and requires enzymatic assistance (flippase) to occur?

- A) Lateral diffusion of lipids
- B) Rotational diffusion of proteins
- C) Transbilayer ("flip-flop") diffusion of lipids**
- D) Flexing of hydrocarbon tails

Q7 Which of the following combination of statements are true

- 1) Be it for positive or negative surface charge density, at any far point x, the surface potential approaches to zero.
 - 2) Only for positive surface charge density, at any far point x, the surface potential approaches to zero.
 - 3) Neither for positive nor for negative surface charge density, at any far point x, the surface potential approaches zero.
 - 4) Negative surface charge potential tends to reach more negative value at far point x.
- A) Only 1 & 4 are correct
 - B) Only 2 is correct.
 - C) Except 3, all statements are incorrect.
 - D) Except 1, all statements are incorrect.**

Q8 The flow of ions across the membrane _____, when the membrane potential reaches _____ potential.

- A) Ceases & Donnan
- B) Starts & Donnan
- C) Ceases & Nernst**
- D) Starts & Nernst

Q9 _____ A _____ includes _____ B _____, but not vice versa.

- A) B = Stern Model, A = Gouy Chapman Model
- B) B = Gouy Chapman Model, A = Stern Model**
- C) B = Helmholtz Model, A = Nernst Equation
- D) B = Donnan Model, A = Stern Model

Q10 Which limitation of the Helmholtz double layer model does the Gouy-Chapman model address?

- A) It accounts for the hydration shell of ions.
- B) It accounts for the disordering effect of thermal motion on ions.**
- C) It assumes ions are point charges.
- D) It ignores the specific adsorption of ions.

Section B

[Marks: 6 X 1]

Q11. _____ is a measure of how far the effect of surface charges may be felt into the bulk of the solution. (Answer: **Debye Length**)

Q12. From the point of view of an ION, the shape of the membrane layer looks like _____.
(Ans: **Straight/tangent/plane**)

Q13 Increasing the ionic strength (salt concentration) of the solution causes the Debye Length to _____, effectively compressing the double layer. (Ans: **Decrease**)

Q14. If the absolute potential inside the plasma membrane (cytosol) is -60 mV and the absolute potential outside (extracellular fluid) is measured at +10 mV, the resulting Transmembrane Potential of plasma membrane is _____ mV. (Ans: **-70 mV (inside negative, or in-out)**)

Q15. When membrane-impermeant charged macromolecules are present on one or both sides of the membrane (along with membrane-diffusible ions), then a _____ potential is developed. (Ans: **Donnan**)

Q16. At the Debye length (d_D), the electrical potential in the diffuse layer decreases to a level _____ of the Stern potential. (Ans: **1/e th**)

Section C

[Marks: 2 X 2]

Q17. Consider the transport of K^+ ion from the intracellular side (suffix “i”) to the extracellular side (suffix “e”) (or vice versa) across the plasma membrane. At equilibrium, if the concentration of K^+ ion

across the membrane at 25°C are $[K^+]_e = 4 \text{ mM}$, $[K^+]_i = 140 \text{ mM}$, then calculate the Nernst Equilibrium Potential for K^+ ion across the membrane.

Ans: Formula to follow: $\Delta G_t = 0 = RT \ln([K^+]_e/[K^+]_i) - ZF \Delta \psi$ (at Nernst Equilibrium Potential for K^+ ion)

- **$\Delta \psi$ convention inside negative**
- **Ratio:** $4/140 \cong 0.0286$
- **Log:** $\log_{10}(0.0286) \cong -1.544$
- **Calculation:** $59 \text{ mV} \times -1.544 \cong \textbf{-91.1mV}$ (rounds to -91 mV).

Q18. Write the ionic strength of 1 mM aqueous solutions of the following strong electrolytes and arrange them in increasing order of Debye length for a charged planar surface.

- A) $\text{Na}_3(\text{PO}_4)$
- B) NaCl
- C) CaCO_3

Ans. A) 6 mM, B) 1 mM, C) 4 mM

(Debye length $B > C > A$)